

Supplementary Material: Documented AI Agent Security Incidents (2024–2025)

This supplement catalogs six documented incidents of AI agent security failures in production systems, retrospectively analyzed through the CIF-AD-OODA framework. Each incident is mapped to the universal attack pattern taxonomy *(sec:attack_patterns)* and the relevant CIF defense mechanism that would have prevented the incident.

Incident Catalog

S2.1 Arup Deepfake Video Conference Fraud (February 2024)

A finance employee at the multinational engineering firm Arup was deceived by a deepfake video conference in which AI-generated replicas of senior executives instructed the transfer of \$25.6 million across 15 transactions. The deepfakes were sufficiently convincing that the employee overrode standard verification procedures, treating the fabricated executive presence as authentic authorization.

CIF-AD-OODA Analysis. The attack constitutes a **Context Boundary Violation**: the boundary between verified identity